

Arnav Surve

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EDUCATION

California State University - Sacramento <i>Bachelor of Science, Computer Science</i>	Sacramento, CA Aug. 2024 – May 2026
Purdue University <i>Bachelor of Science, Cybersecurity – Dean's List, Fall 2023</i>	West Lafayette, IN Aug. 2022 – May 2024

SKILLS

Languages: Go, Python, TypeScript, JavaScript, SQL, PostgreSQL, Java, HTML/CSS, YAML, JSON, Bash
Frameworks: React, Node.js, FastAPI, Express.js, Echo, GORM, GraphQL, Prisma, Model Context Protocol
Cloud/DevOps: AWS, Lambda, Glue, RDS, ECR, EKS, Kubernetes, Docker, Terraform, Git, GitHub Actions
Data/Blockchain: PostgreSQL, Redis, ETL, VSAM, Blockchain, Smart Contracts, Tezos, NFTs, Web3
Practices: CI/CD, Microservices, REST APIs, Agile/Scrum, DevOps, Monitoring, IaC

EXPERIENCE

Software Engineer Intern <i>California Department of Motor Vehicles</i>	March 2025 – Present Sacramento, CA
- Architected and deployed end-to-end ETL pipeline processing FTP drops of mainframe VSAM dataset exports for vehicle registrations and driver licenses, successfully migrating 197 million records to production RDS PostgreSQL database serving California Highway Patrol, FBI, and other federal law-enforcement agencies. - Developed comprehensive data processing solution using AWS Lambda and Glue for initial bulk load and daily synchronization pipelines handling 200-300k daily records with upsert/delete operations. - Implemented entire infrastructure as code using Terraform, managing deployment lifecycle across dev, UAT, and production environments while supporting comprehensive UAT testing throughout development cycle. - Refactored and deployed distributed microservices blockchain system using TypeScript monorepo with 5 interconnected Tezos smart contracts (FA2 NFTs), implementing multitenant role-based access control and chain bridge worker pools for database-blockchain synchronization. - Deployed self-hosted private Tezos blockchain node infrastructure on Kubernetes using ECR and EKS, orchestrating containerized microservices with production-ready CI/CD pipelines and comprehensive monitoring.	

IT Infrastructure Intern <i>MITER Brands – Milgard Manufacturing</i>	June 2024 – Aug. 2024 Tacoma, WA
- Implemented APM pipeline optimizations in Site24x7 and New Relic, reducing alert redundancy by 60% across 6 teams and 2 business units. - Developed REST API endpoint monitors identifying critical discrepancies, resulting in an average 20% latency reduction due to increased observability. - Automated Active Directory compliance audits using PowerShell scripting, improving operational efficiency by 20%.	

Software Engineer <i>Purdue University Office of Engagement</i>	Aug. 2022 – May 2023 West Lafayette, IN
- Delivered full-stack research grant application platform using Agile Scrum methodology in 10-person development team. - Built RESTful API with Express.js and PostgreSQL using Prisma ORM, implementing federated SSO authentication. - Containerized application with Docker for deployment to university server cluster infrastructure.	

PROJECTS

Grace Programming Language <i>Go, Compiler Design, Type Systems, Code Generation</i>	2024
- Built statically-typed programming language (8,407 LOC) that compiles to readable COBOL, featuring complete compiler toolchain with lexer, recursive descent parser using Pratt parsing, and comprehensive AST for semantic analysis. - Designed novel width-aware type system with compile-time COBOL PIC clause validation, automatic width inference for arithmetic expressions, and overflow detection for mainframe compatibility. - Implemented sophisticated code generation translating modern constructs (procedures, records, type inference) into structured COBOL with automatic temporary variable management and DECLARATIVES sections.	
Grace Infrastructure Platform <i>Go, Zowe CLI, YAML, DAG Execution, Mainframe Integration</i>	2024
- Engineered hybrid mainframe-cloud orchestration platform (8,366 LOC) modernizing COBOL workflows through declarative YAML configuration and Infrastructure-as-Code principles. - Built sophisticated DAG execution engine with concurrent job processing, dependency resolution, and intelligent state management for complex multi-step mainframe-cloud workflows. - Implemented extensible plugin architecture supporting multiple job handlers (compile, linkedit, execute, shell) with automatic cleanup of temporary mainframe datasets and comprehensive error handling.	